ANATOMY OF THE CELL

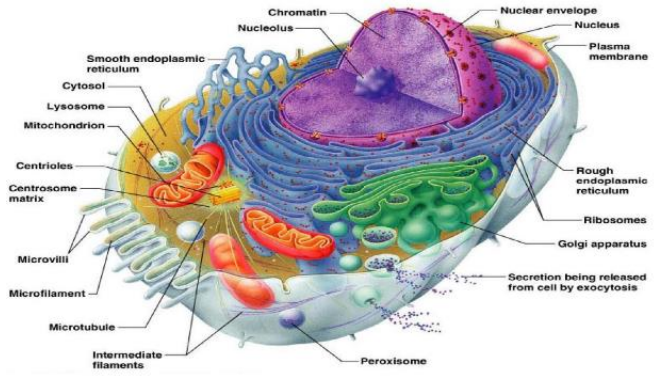
CELLS

- The structural units of all living things
- All cells share general structures but not all cells are the same
- All cells have 3 main regions – nucleus, plasma membrane, and cytoplasm.

CELL THEORY

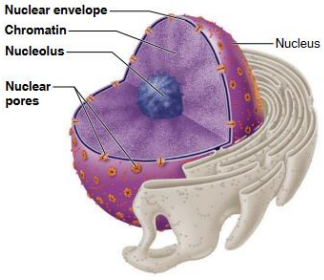
- A cell is the basic structural and functional unit of living organisms.
- The activity of an organism depends on the collective activities of the cell.
    - **PRINCIPLE OF COMPLEMENTARITY** – the activities of cells are dictated by their structure (anatomy) and function (physiology).
    - Continuity of life has a cellular basis.

STRUCTURE OF A GENERALIZED CELL



THE NUCLEUS

- The control center of the cell
- Contains genetic material – the **DNA**, which is a blueprint that contains all the instructions needed for building the whole body.
    - DNA has genes, which carry the instructions for building proteins.
    - DNA is necessary for cell reproduction.
- The shape of the nucleus usually conforms to the shape of the cell.



    - EX. If the cell is elongated, the nucleus is usually elongated as well.
- Necessary for cell division and replication
- It has 3 regions – nuclear membrane (nuclear envelope, nucleolus, and chromatin.

NUCLEAR MEMBRANE

- Is a double membrane barrier
    - Between the two membranes is a fluid-filled “moat” or space
- Allows some, but not all, substances to pass through.
- It encloses a jellylike fluid called **NUCLEOPLASM.**
- **NUCLEAR PORES**
    - An opening after the two layers of the nuclear envelope fuse.

NUCLEOLUS

- The nucleus contains one or more small, dark-staining, essentially round bodies called nucleoli.
- Ribosome production site

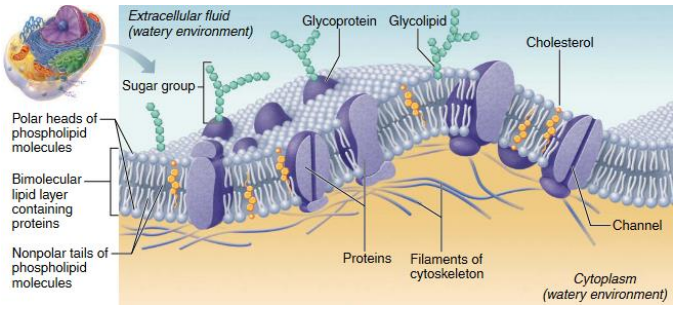
CHROMATIN

- When a cell is ***NOT*** dividing, its DNA is carefully wound around proteins called histones to form a loose network of beads on a string called chromatin.
    - It is scattered throughout the nucleus.
- When a cell is ***DIVIDING*** to form two daughter cells, the chromatin threads coil and condense to form ***CHROMOSOMES***.

THE PLASMA MEMBRANE

- A fragile, transparent barrier that contains the cell contents and separates them from the surrounding environment.
- Made of phospholipid bilayer and contains proteins, cholesterol, and glycoproteins.
    - The proteins allow cell adhesion, cellular transport and communication.
- It determines which substances enter and leave the cells.

STRUCTURE OF THE PLASMA MEMBRANE



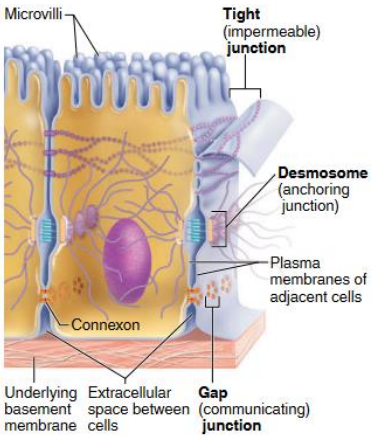
THE FLUID MOSAIC MODEL

- The structure of the plasma membrane consists of 2 phospholipid (fat) layers arranged tail to tail with cholesterol and floating proteins scattered among them.

- Some phospholipids may also have sugar groups attached, forming glycolipids.
- The proteins, some of which are free to move and bob in the lipid layer, form a constantly changing pattern or mosaic.
- The olive-oil like phospholipid bilayer forms the basic fabric of the membrane.
- The polar heads of the lollipop-shaped phospholipid molecules are ***HYDROPHILIC*** (water loving).
- Their nonpolar fatty acid tails are ***HYDROPHOBIC*** (water fearing) and line up in the center (interior) of the membrane.
- The hydrophobic makeup of the membrane interior makes the plasma membrane relatively impermeable to most water-soluble molecules.

PLASMA MEMBRANE SPECIALIZATIONS – CELL MEMBRANE JUNCTIONS: TYPES OF JUNCTIONS

1. **TIGHT JUNCTIONS**
   - Impermeable junctions that encircle the cells and bind them together into leak proof sheets.
   - Adjacent plasma membranes fuse together tightly like a zipper and prevent substances from passing through the extracellular space between cells.
   - A barrier to the movement of molecules across membranes.
2. **DESMOSOMES**
   - Anchoring junctions scattered like rivets (tool ni siya na murag screw) along the sides of adjacent cells.
   - They prevent cells subjected to mechanical stress from being pulled apart.
   - Staples cells in between for tight intercellular adhesion.
3. **GAP JUNCTIONS**
   - Communicating junctions that are commonly found in the heart and between embryonic cells.
   - A bridge for sharing molecules for rapid intercellular communication.



MICROVILLI – a finger-like projection of a tissue, cell, or cell organelle that increases surface for absorption.

THE CYTOPLASM

- The cellular material outside the nucleus and inside the plasma membrane.
- It is the site of most cellular activities – "factory floor" of the cell.
- It has three major components – cytosol, inclusions, and organelles.

CYTOSOL

- A semitransparent fluid that suspends the other elements.

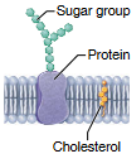
INCLUSIONS

- Chemical substances that may or may not be present, depending on the specific cell type.
- In short, non-functioning units.

ORGANELLES

- This is the metabolic machinery of the cell.
- Each type of organelle is specialized to carry out a specific function for the cell as a whole.

CELL PARTS, FUNCTIONS, AND STRUCTURE

Cell part*	Structure	Functions
Plasma Membrane (Figure 3.2)		
	Membrane made of a double layer of lipids (phospholipids, cholesterol, and so on) with proteins embedded within. Most externally facing proteins and some lipids have attached sugar groups.	Serves as an external cell barrier and acts in transport of substances into or out of the cell. Maintains an electrical condition that is essential for functioning of excitable cells. Externally facing proteins act as receptors (for hormones, neurotransmitters, and so on), transport proteins, and in cell-to-cell recognition.
Cytoplasm	Cellular region between the nuclear and plasma membranes. Consists of fluid **cytosol** containing dissolved solutes, **organelles** (the metabolic machinery of the cytoplasm), and **inclusions** (stored nutrients, secretory products, pigment granules).	
Organelles		
• Mitochondria (Figure 3.4) 	Rodlike, double-membrane structures; inner membrane folded into projections called cristae.	Site of aerobic respiration (the "burning" of glucose) and ATP synthesis; powerhouse of the cell.
• Ribosomes (Figures 3.4, 3.5) 	Dense particles consisting of two subunits, each composed of ribosomal RNA and protein. Free or attached to rough endoplasmic reticulum.	The sites of protein synthesis.
• Rough endoplasmic reticulum (Figures 3.4, 3.5) 	Membranous system enclosing a cavity, the tunnel, and coiling through the cytoplasm. Externally studded with ribosomes.	Sugar groups are attached to proteins within the tunnels. Proteins are bound in vesicles for transport to the Golgi apparatus and other sites. External face synthesizes phospholipids.
• Smooth endoplasmic reticulum (Figure 3.4) 	Membranous system of tunnels and sacs; free of ribosomes.	Site of lipid and steroid (cholesterol) synthesis, lipid metabolism, and drug detoxification.
• Golgi apparatus (Figures 3.4, 3.6) 	A stack of flattened membranes and associated vesicles close to the ER.	Packages, modifies, and segregates proteins for secretion from the cell, inclusion in lysosomes, or incorporation into the plasma membrane.

Cell part*	Structure	Functions
• Peroxisomes (not shown) 	Membranous sacs of oxidase and catalase enzymes.	The enzymes detoxify a number of toxic substances such as free radicals. The most important enzyme, catalase, breaks down hydrogen peroxide.
• Lysosomes (Figures 3.4, 3.6) 	Membranous sacs containing acid hydrolases (powerful digestive enzymes).	Sites of intracellular digestion. The "stomach" of the cell.
• Microtubules (Figure 3.7) 	Cylindrical structures made of tubulin proteins.	Support the cell and give it shape. Involved in intracellular and cellular movements. Form centrioles and cilia and flagella, if present.
• Microfilaments (Figure 3.7)	Fine filaments composed of the protein actin.	Involved in muscle contraction and other types of intracellular movement; help form the cell cytoskeleton and microvilli, if present.
• Intermediate filaments (Figure 3.7) 	Protein fibers; composition varies.	The stable cytoskeletal elements; resist mechanical forces acting on the cell.
• Centrioles (Figure 3.4) 	Paired cylindrical bodies, each composed of nine triplets of microtubules.	Organize a microtubule network during mitosis (cell division) to form the spindle and asters. Form the bases of cilia and flagella.
Inclusions	Varied; includes stored nutrients such as lipid droplets and glycogen granules, protein crystals, pigment granules.	Storage for nutrients, wastes, and cell products.
Nucleus (Figures 3.1, 3.4)		
	Largest organelle. Surrounded by the nuclear envelope; contains fluid nucleoplasm, nucleoli, and chromatin.	Control center of the cell; responsible for transmitting genetic information and providing the instructions for protein synthesis.
• Nuclear envelope (Figure 3.1) 	Double-membrane structure pierced by large pores. Outer membrane continuous with the endoplasmic reticulum.	Separates the nucleoplasm from the cytoplasm and regulates passage of substances to and from the nucleus.
• Nucleolus (Figure 3.1) 	Dense spherical (non-membrane-bounded) bodies, composed of ribosomal RNA and proteins.	Site of ribosome subunit manufacture.
• Chromatin (Figure 3.1) 	Granular, threadlike material composed of DNA and histone proteins. "Beads on a string."	DNA constitutes the genes, which carry instructions for building proteins.

CELL PHYSIOLOGY

PLASMA MEMBRANE

- Is a selectively permeable barrier – meaning that a barrier allows some substances to pass through it while excluding others.
- Transport through the plasma membrane occurs in two basic methods:
    - **PASSIVE TRANSPORT** – no energy is required
    - **ACTIVE TRANSPORT** – cell must provide metabolic energy

PASSIVE TRANSPORT PROCESSES

1. **DIFFUSION**
    - The particles tend to distribute themselves evenly within a solution.
    - Movement is from high concentration to low concentration, or down a concentration gradient.

TYPES OF DIFFUSION

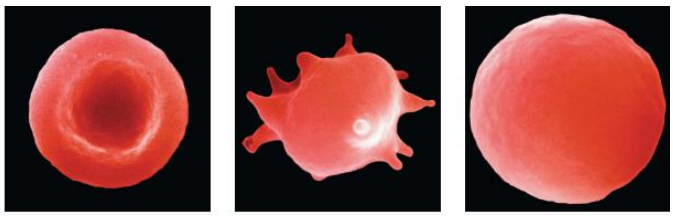
- **SIMPLE DIFFUSION** – unassisted process; solutes are lipid-soluble materials or small enough to pass through the membrane pores.
- **OSMOSIS** – simple diffusion of water; highly polar water easily crosses the plasma membrane.
- **FACILITATED DIFFUSION** – substances require a protein channel or carrier for passive transport.

2. **FILTRATION**
    - Water and solutes are forced through a membrane by fluid, or hydrostatic pressure.
    - Solute-containing fluid is pushed from a high pressure area to a lower pressure area.
    - In kidneys, water and small solute filter out of the capillaries into the kidney because blood pressure in the capillaries is greater than the fluid pressure in the tubules.

CELLULAR TONICS

OSMOTIC PRESSURE

- The tendency of a solution to hold water or pull water into it.
- The higher the solute concentration, the greater the tendency of water to move into the solution.



(a) RBC in isotonic solution (b) RBC in hypertonic solution (c) RBC in hypotonic solution

HYPERTONIC

- Has a higher concentration of solutes than the fluid inside the cell.

ISOTONIC

- Has the same concentration of solutes as the fluid found inside the cell.

HYPOTONIC

- Has a lower concentration of solutes than the fluid inside the cell.

ACTIVE TRANSPORT PROCESSES

- Transport substances that are unable to pass by diffusion because:
  - They may be too large
  - They may not be able to dissolve in the fat core of the membrane.
  - They may have to move against a concentration gradient.
- Two common forms of active transport:
  1. **SOLUTE PUMPING**
     - Amino acids, some sugars and ions are transported by solute pumps
     - ATP energizes protein carriers, and in most cases, moves substances against concentration gradients.
  2. **BULK TRANSPORT**
     - ***Endocytosis*** – into the cell
       - ***PINOCYTOSIS*** – taking in liquid cell; cell drinking
       - ***PHAGOCYTOSIS*** – taking in particles; cell eating
     - ***Exocytosis*** – out of the cell

CELL DIVISION

CELL LIFE CYCLE

- The series of changes a cell goes through from the time it is formed until it divides.

2 MAJOR PERIODS

1. **INTERPHASE**
   - The cell grows and carries on its usual metabolic activities.
   - Metabolic phase
   - Longer phase of the cell cycle
   - During interphase, the cell is very active and is preparing for cell division
2. **CELL DIVISION**
   - Cell replicates/reproduces itself.
   - Its function is to produce more cells for growth and repair processes.

DNA REPLICATION

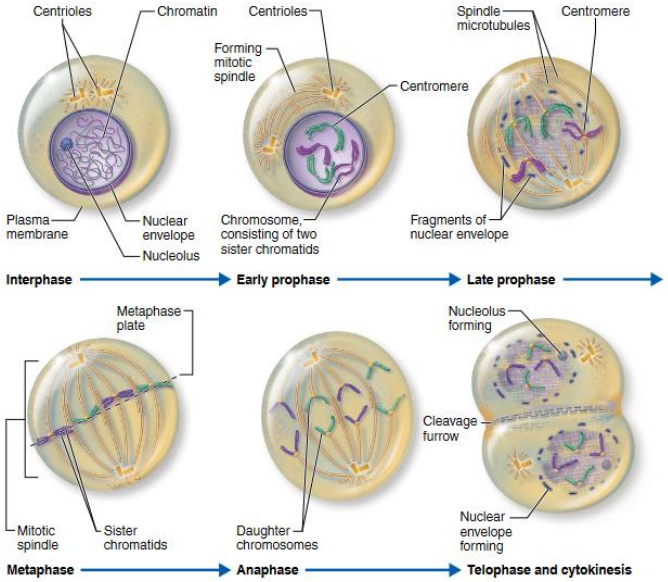
- Genetic material that is duplicates and prepares a cell for division into two cells.
- Occurs at the end of the interphase.
- DNA uncoils and each side serves as a template.

EVENTS OF CELL DIVISION

MITOSIS

- The process of dividing a nucleus into 2 daughter nuclei with exactly the same genes as the mother nucleus.

STAGE OF MITOSIS



1. **INTERPHASE**
   - No cell division occurs
   - The cell carries out normal metabolic activity and growth.
2. **PROPHASE (LATE AND EARLY)**
   - Frist part of cell division
   - Centromeres migrate to the poles.
3. **METAPHASE**
   - Spindle from centromeres are attached to chromosomes that are aligned in the center of the cell.
4. **ANAPHASE**
   - Daughter chromosomes are pulled toward the poles
   - The cells begin to elongate
5. **TELOPHASE AND CYTOKINESIS**
   - Daughter nuclei begin forming a cleavage furrow (for cell division) and begins to form.

PROTEIN SYNTHESIS

- Protein synthesis is accomplished with the DNA blueprints (genes) and with the help of RNA.
  - **GENE** – a DNA segment that carries the information for building one protein.
- DNA's information is encoded in the sequence of bases. Each sequence of 3 bases calls for a particular ***amino acid*** – the building blocks of proteins and are joined during protein synthesis.

THE ROLE OF RNA

- There are three varieties of RNA that play a special role in protein synthesis:

rRNA	**Ribosomal RNA**; helps form the ribosoms
mRNA	**Messenger RNA**; carries the message containing instructions for protein synthesis from the DNA (gene) in the nucleus to the ribosomes in the cytoplasm.
tRNA	**Transfer RNA**; escorts amino acids to the ribosome.

THE PROCESS OF PROTEIN SYNTHESIS

1. **TRANSCRIPTION**
    - Involves the transfer of information from the sequence of bases in a DNA gene into the complementary sequence of mRNA by an enzyme.
    - ***DNA is the template for transcription, mRNA is the product.***
    - Each three-base sequence specifying a particular amino acid on the DNA gene is called a triplet, and the corresponding three-base sequences on mRNA are called codons. The form is different, but the same information is being conveyed.
        - Thus, if the (partial) sequence of DNA triplets is AAT-CGT-TCG, RNA base-pairing rules (A:U, G:C) tell us that the corresponding codons on mRNA would be UUA-GCA-AGC.
2. **TRANSLATION**
    - The language of nucleic acids (base sequence) is translated into the language of proteins (amino acid sequence).
    - It occurs in the cytoplasm and involves the 3 major varieties of RNA.